The Free Energy Principle with Dr. Maxwell Ramstead | Talk of Today Podcast

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my name is Maxwell ramstad I'm currently a postdoctoral fellow at the Jewish General Hospital in Montreal and I'll be uh I'll be staying at McGill for a for a while yeah I'm also affiliated with McGill University which is the main University affiliation of the hospital and I'll be a postdoc in the department of psychiatry at McGill starting next September yes so basically my background is in philosophy and cognitive science with some additional training in computational neuroscience yeah so currently I work on um basically multi scale active inference so multi-state multi scale extensions of applications of the free energy principle to basically like systems beyond the brain so like systems well within and beyond the brain right so what I'm interested in developing is kind of a multiscale approach to cognitive systems and that's that's how I got into the free energy principle originally so I started off it more interested in this this field called neuro phenomenology right so like the the kind of so phenomenology is a philosophical discipline originally and it's about the kind of rigorous description of first-person experience so neuro phenomenology was a was a project it's still ongoing in part the project to bring together this rigorous description of first-person lived experience with you know the neurosciences and other other sciences that might tell us something about how experience comes about like first-person lived experience yes I was working on that for a while and I I was looking for a way to talk about what I think is like a really central intuition which is that like all of the different levels of description of reality are like they have some in a way that that's like like there's there's an equal dignity to all these different descriptive levels so like you know we are physical systems right we're made of atoms so the laws of you know the regularities that pertain to the physical world apply to us but we're also biological creatures we're made of cells right and and we have a psychology right and we conglomerate in social groups that have you know social and cultural scripts and so we also have like a rich you know infinitely rama fiying first-person experience and somehow all of these things are true at the same time right so how can we develop a framework that addresses all of these levels yeah it would equal dignity you know with take taking each seriously and so you know around 2014 I was looking for a kind of glue that would allow me to kind of talk rigorously about all exactly about all these levels at the same time and I I met Carl Tristan basically and so the rest is kind of history it really really started me off on my current path and yeah right now I work I work with Carl very closely and who is Co alfredston yeah but what's his what's his significance in this picture yeah right so Kristen is I think right now the the most cited living neuroscientist he's he's known in the field I mean for several different reasons that what is is probably most lasting contribution is well so far has been the development of basically the statistical packages that are allow us to perform neuroimaging analysis and in that in the fMRI modality so in the 90s he basically invented statistical parametric mapping yeah which is the the main tech used in neuroscience so is that like taking the data and it's like a something out of that that enables you to like interpret the data or actually make sense of it like one thing but actually be able to the whole framework that I'm going to be explaining it's very complicated it looks very complicated but conceptually it's very simple you can explain it with two circles and a line I like this thing all right so the two circles are data right that you have so observations states that you can observe or measure and latent or hidden states that caused your data right so I mean to just start with the beginning I guess so you know are you familiar with fMRI neuro imaging so yeah full mag magnet yeah so basically an fMRI machine is a big magnet and you put people in this big machine and it measures it measures a if the signal that it measures is called the bold signal it's the blood oxygenation level dependent signal so this is something to keep in mind whenever you see neuro studies in general typically you're not directly measuring brain activity so these heat maps of brain activity that you see are not directly measuring brain activity you're always measuring a proxy for brain activity right so in an fMRI that proxy is is the bold signal this blood basically one of the metabolic byproducts of neural activity has these interesting magnetic properties that we can pick up using these big magnets right so you put someone in a big magnet you and then you scan their brain using the magnet and essentially you're picking up a a byproduct of oxygen consumption by the brain so then that's your data and then what you do using these statistical techniques that Carl and and colleagues develop is work backwards from this data to basically the the most probable set of hidden causal factors that generated your data yeah so he's he came up with this in the 90s and continued extending these methods so I mean the idea here is always sort of the same right so with what you're doing is you're writing down different alternative causal structures or models of the causal structure of the process that generated your data right so these techniques have been extended recently and in principle you can model any kind of data using these kind of techniques so recently Carl is used like the newest version of these frameworks dynamical causal modeling to model the spread of COBIT as it happens so in this case like the the the data that you're using is the raw numbers like so the total number of cases total number of deaths and the product the causal process that you're estimating is stuff like you know is there confinement what is the population size like all of these different parameters in your model that you're estimating given given your data so that's that's basically like the the kind of general framework that he developed active inference per se gets into the game when you take the same kind of framework and say okay what does it look like when the data that I'm trying to explain isn't like neuroimaging data it's the actual sensory data that I'm generating as an organism like yeah so that's that's active inference in a nutshell it's it's a framework to that extends these modeling techniques in a way that it's sort of like what would it look like if the the organisms themselves leveraged these strategies to effectively infer the latent causes of their sensory states so active inference is something that organisms do but it's not the technique used to measure the sensory data but it's something that that organisms do like we actively framework is correct so you can think of the active inference framework as this mathematical theoretical apparatus we're applying to biological systems to understand them but on the assumption that the framework is correct yeah active inference is something that living systems engaged in basically this kind of so active inference essentially has two sides right on the one hand you're you're trying to infer which which kind of constellation of causal factors is currently causing my my sensory phenomenology right so yeah um that's on the one hand and on the other and this is like a crucial point active inference is about its so this is what makes it different from like older Bayesian theories like predictive coding for example it's a theory of active sampling of the world so it's not merely a question of like Basel optimal perception it's a question of how do I act such that I bring about my preferred observations it's just like this this tie between the entity acting within the environment taking its sensory input I try to understand or identifying what's actually causing what so it's like this interplay between sensory experience and action and that's where I essentially yeah yeah yeah it's a it's a formal model of the sensory motor loop is another way to think about it okay cool cool and serve the free-energy principle so come first and what's the relationship between there's a few ways I want to go about this conversation because I want to talk about cognition and information for assessing and I also want to talk about I mean the purpose of this podcast is really to find out what the hell the free-energy principle is and like what are its applications and how does it actually work because in other podcasts like I came across the idea and I'm just being like very excited about it and like a few of the podcasts that I've had I'm just like oh yeah this thing kind of applies here I just kind of give a half-baked description of it which is one of the reasons why I want to talk to you just I actually know what the hell I'm talking about so do you think would make more sense to to kind of discuss cognition and information processing and you know dive deeper into active inference or just start with the free energy principle and go from there what do you think maybe maybe I can I can show you a few slides yeah if we can discuss it can we describe the slides as well just because it's absolutely just to give you I mean it's really just to be able to see it at the same time so that's for those of you who will be listening the podcast without any visual support it's really just the two circles and the line thing that I was discussing earlier but it's it's good it's it's often useful to to see I'm a visual person anyway I don't know about you know I find it helpful okay so yeah if anyone wants to check out the the YouTube video or just photos so like all the photos and like you're gonna enable screen sharing long way oh yeah yep oh yeah right all these privacy features all right so this is a kind of cartoon tutorial so this is what I was saying earlier right two circles on the line so what you have is you have data right observable states that you have access to so again like if you're working in fMRI for example your data is your bold signal and what we're trying to do is basically go the opposite direction here right so causality flows from hidden States to data you can also think of prediction is going in this way right so like if you have a model of how some phenomenon it works you can predict that you'll be generating this kind of data rather than that kind of data right that's sort of the point of being able to use these predictive models I mean you see this a lot with the koban modeling right people have a model of how the process the the propagation process happens and so that they kind of can project the kinds of well data that they will be registering in the future right so causality or prediction moves in this direction and inference moves in that direction so these hidden states or latent causes that are generating our data what we do is we write a model right of the basically the relations between the factors such that they generate the the outcome of interest rate and the variational free-energy I think the easiest way to think about it is that is that it's a it's a measure or more precisely an approximation of how much evidence we have for a model given the data right so like you can think of I don't know like say two or three alternative models of how some data was generated right and the free energy is basically a model of how what's Rea a measure of how well each of the models explains the data like so how much of the variance for example your what do you say explains you don't mean do you mean actually explains why it happens or predicts what's going to happen like is it is it it's not an explanation of how this causes predictive explanation if we're gonna if we're gonna like draw from the philosophical literature and talk about like models of explanation so if you're thinking about mechanistic explanation that there are a few missing elements here but basically like which which you're doing is saying well given the data that I have which of these models is most probable given yeah like given the observations that I've made which of these models is most probable and the free-energy is is essentially a metric or a measure of that that the amount of evidence that the data makes available for each of the different models what sorry about your continuing yeah so I mean this is the fMRI setup that I was discussing earlier right so you have a bold signal you have hidden causes which in this case are the underlying neurobiological activity and what you're doing is essentially constructing alternative models of for example brain structure or connectivity right and then scoring how well each of those models alternative models explains the data that you're generating with the assumption that like the the model that so it's a kind of it's an abductor kind of or inference to the best explanation kind of framework for what you're saying is well you know the the model that most probably describes the actual causal process that generated my data is the one associated with the most evidence yeah so in in the what so transposing exactly the same reasoning to the brain you have this kind of setup active inference per se where the data that I'm trying to explain is the the sensory states of my organism right so like the states in my retina right the states is like my my auditory organs like my cochlea and all that like the that my the state of mice you know receptor cells in my skin and the model here isn't the model of I mean at some stage it might be but like the the the model is ultimately a model of the agent acting in the world since our actions are the main driver of our sensory phenomenology yeah so in a nutshell that's kind of the the framework and in in all these cases free energy is basically a measure of evidence yeah so I've got I had two questions one I think I'll just put to you my layman's understanding of it I want you to tell me if I'm right or wrong and then after that I want to know why is it called wise is free energy the measure like why that term so the way I sort of see it and I think I've described it to friends is that organisms and organisms have a model of they have an internal model of themselves and the world in which they inhabit and they try to enact they try to act in the world in such a way to minimize uncertainty or surprise and this is kind of the the mechanism by which it happens so this is like the description of it right I would knew once that a bit and say that the free energy principle tells you not just that you have a model but that you are a model right so in the sense that like it's not like you carry around I mean to a certain extent you do right like there is something like a model in the brain right like it's not it's not quite that I mean what we would prefer to say I think is that like the brain embodies parameters of a model but the whole model is something like the brains inference activity so look to think of it this way like this this issue of model has to do with predictability right so what does it mean to say that I'm a model of my environment well so look we're having this conversation I'm wearing a sweater right so you can tell from my wearing a sweater or you can predict from my wearing a sweater that it's it must be 20 degrees Celsius roughly in my room maybe a bit less maybe 15 right because it's a sweater it's not like a t-shirt so so there are these like systemic relationships between the organisms phenotype right and the state of the environment these are predictive relationships essentially so that's the sense in which you can say that the organism is a statistical model of its environment that's that so like the physiology of the fish tells you something about the kind of environment that it lives in so the fish is a statistical model of an underwater environment right so and and the fish's phenotype it becomes extremely unlikely on land right just like I would be very surprised to find myself in water all of a sudden so that's the intuitive notion there it's not just that and this is where it departs from older classical like cognitivist theories to an extent I mean in my view that kind of indicates you know classical cognitivist ideas through through the lens of these new cool like embodied and active kind of stories yeah like on the classical story like I I reconstruct an internal model in my head and that guides my action so that that story is still here in part I mean you know clearly as human organisms we do construct these explicit internal models that we manipulate and they're the same kinds of models but there's like a broader notion at play here it's not just that I have a model is that I am a model and I really okay okay and you said inactive and embodied and like I've jumped on Twitter of maybe last year and I just keep saying you know in activism and embodied cognition popping up more and more and I spoke to Lisa Feldman Barrett a while back she's great yeah yeah yeah and this kind of breeding her book how emotions are made I think it's gonna just blew my mind complete change my worldview and I guess very powerful point of view that she's defending I mean I'm very very very friendly to the kind of constructed this a view of emotion that's being pushed by that group it's very yeah and I guess just for those who aren't familiar and Maxwell if I you know don't do it don't give do it justice please jump in but what I took away with from it was like we have all these underlying bodily sensations that we interpret and then give meaning to in a way and you know they could be we may feel ignoring in the stomach and we may interpret that as hunger or it could actually be anxiety and like we directly the emotion from that and something and this ties in to what we're talking about here if you're active inference that emotions in a way [Music] the I think yeah emotions are like an emergent form of prediction and they're like how we at this at the at this scale of the scale of this organism how we predict what's going on the world and then react to it right and like emotions arise based on sometimes the mismatch of our prediction and the the actual results yeah precisely I mean we have a preprint that's in revision right now in neural computation where we make basically that kind of argument which is to say that well myself and our co-authors Caspar Hess who I collaborate with very closely Brian Smith is also on the paper these are all like very cool researchers Kasper is probably the person I've been collaborating with the most recently he's he has a background in computational astrophysics oh I am still sharing my screen sorry yeah he so the paper is called deeply-felt effect yeah so also on the paper is a Thomas Parr who is a UH like a it's basically a genius I mean Thomas is like one of the most impressive scholars that I that I know he's Carl's right-hand man basically yeah so and Myka Allen is also on the paper and myself Mike has a cool intro interoception yeah yeah he does very cool stuff as well you might want to check out his his research but yeah so Casper Ryan Thomas myself and Carl are on this paper and what were essentially arguing is that you can think of basically you can think of a motion by unpacking it formally in terms of basically state state inference about your state inferences so like higher level inferences about how well you're doing with your lower level inference stuff so we're sort of saying that emotion arises along with kind of implicit metacognition but as soon as you start to monitor yourself and to assess how well you're doing implicitly you're you're you're you're moving towards like emotional types of inference we're like yeah so I mean to keep it simple in a nutshell that's what we're arguing in that day yeah well what's really exciting about this perspective is that it brings like you know we used to some people think of emotions as just like these useless relics so Adam Ally and past right there they have absolutely no function and that if we were rational they would do away with emotion I guess from this perspective it's like it's vitally important to how we interact with the world and it's like it's this you could say like is this compass of sorts that has emerged over time and it's actually highly attuned to well out ourselves and our place in the world and that without these things without these feelings that we would just be bereft of any navigational means we just yeah exactly a hundred percent and I mean so the the we were talking about like embodied and inactive so like all these breakdown there's if you could break those down in some way typically which we'll hear about is the four e a approach right so the four es are extended embedded embodied and inactive and the a is effective so we just covered the a right and so all of these perspectives are meant to be correctives or criticisms of the traditional kind of cognitivist framework susan hardly calls this the cognitive sandwich right where you have perception cognition and action right in that sequence so perception is basically like taking in passively the world cognition is creating an internal model that you used to then that you manipulate to make sense of the world and then use the plan action and then action is seen as sort of the motor output of this kind of process so the cognitive sandwich right yeah so the the fourier approach says this is wrong for several reasons so that we just discussed the the a reason that it's wrong right cognition isn't just like this kind of flat calculating process right it's it essentially involves you know self-monitoring self appraisal you know the emotional reactions right so the the these are also correctives of this cognitive this perspective right so the I guess the two E's that kind of come together embedded and extended which are basically like one more and one less radical version of the same claim so to illustrate it I'll give you an anecdote Richard Feynman was discussed the famous physicists right who was involved with the Manhattan Project one of my idols III I love the man he's just amazing so Feynman was talking to this man who had bought the notes that he'd jot where he jotted down like a famous equation like the original kind of notes that where he worked out his equations and he said look you know I have a I have a trace of you know the you're thinking and Feynman responds no what you have is is my thinking you know it's not it's not just like a trace or like a testimony of my thinking I I used those equations to think through them on the paper right so this this comes together under the the broad rubric of embedded and extended cognition so the idea that like cognition is something that is essentially world involving it's not just like it's not just like cognition is something that's in the head and sometimes in some context that spills out into the world in interesting ways it always already involves engagement with the world and then you know the differences in metaphysics like if you want to say that the world literally becomes part of the cognitive system as this kind of makes this kind of cognitive overspill happens like you know if you if you literally want to say that when you are writing an equation on a blackboard the blackboard becomes part of your mind then you're in the extended mind cam yeah so the mind is literally extending yeah yeah you know when it when when possible metaphysical inconvenience of this view though is that it means that like when you're when you're using Google to perform a query for for a few seconds or you know whatever fraction of a second it is like the the servers are actually part of your cognitive processing some people want to avoid that implication and we just want to jump in is mind synonymous or like what's the connection between mind and cognition here because I mean I think if the cognition as information processing whereas mind is very different experience that's that's the nuance in the literature some people I mean I think people more safely talk about extended cognition yeah I think that giggles just got a part of my brain and that you know like my mind isn't Google is a little bit crazy yeah so it is safer indeed from a metaphysical point of view to talk about extended cognition all right rather than the extended mind I think the thesis is used interchangeably like the label for the thesis although you're right like that is a Andy Clark I think discusses the new ones he's one of the people who's who's worked on this a lot his I mean the original the original paper I think is from is different Park controllers yeah it's let me make sure Clark Chalmers the extended mind yeah exactly it's Andy Clark and David Chalmers are the the people who kind of originally proposed at least the contemporary version of this yeah and Andy has this great book supersizing the mind which which kind of takes the idea and runs with it yeah but so I think some people would want to say yeah it literally is the mind that's being extended some people want to say no it's just cognition but you could also think that this is a bit of a wacky idea and want to recognize that like there's there's a situated aspect an ecological aspect of cognition without buying the cognitive process he's actually extend beyond the skull is due to biological do you mean like we are embedded within some environment and that yeah precisely any interest will be embedded mine yeah extended and embedded embedded is like the less radical cousin and extended okay it kind of issues the whole like metaphysics of those the mind extend does it make sense to say the cognitive process you spill out and says well clearly get some kind of ecologically valid behavior in humans cognition needs to be scaffolded or embedded in the right kinds of context so it I mean you know for some people who don't like the radical the radical I'm not one of those I won't I want something more radical personally but yeah that option does exist though in the literature mm-hmm so what how do you think of cognition like what is cognition and how is it different to information processing I mean I'm sure they're very very similar but like when does a form of information processing become cognition so there is the right question I'm not sure I'm not sure anymore I mean because by talking about you know like if through this lens and if I'm writing equations of a blackboard or like typing into my computer or you know interacting with a Wolfram Alpha or something and that's performing its processing information on my own behalf when does that become cognition and if it is cognition then to all my cells because they all process information my immune system is processing information like my hormone is processing information at all times like my individual cells if I had to I I I'm sort of I don't really like these broad concepts like life

Caucasian no but I mean I I suspect that it's the kind of thing that breaks down when you look at like the the kind of border cases yeah yeah I had a a when I was an undergrad like a decade ago like I I I had a philosophy of biology professor who was an elimination ista bout life Frederick Bouchard University of Montreal and that kind of stayed with me I think he says like what's an elimination Estelle it's it's an elimination isten general is someone who thinks that X or Y concept actually doesn't serve much of a purpose anymore and she can be used scientifically so a typical example of elimination in the progress of science is like elemental fire of the ancients right so you know the in in in ancient Greece or whatever you know they there were these ontologies these elemental ontology x' right where like the the universe is like for it for Aristotle like a combination of different elements and you know that there are similar versions of these ontologies like across cultures like you know with different elements like I think that the Chinese elements or something like metal wood fire you know we don't have in the West these as ontologies didn't have metal but whatever you know and so fire under these ontological classifications which were I mean you know they they were sort of scientific at the time I mean I don't know if if it makes sense to project our modern understanding of scientific practice back back on 2 degrees but certainly they were using these concepts explanatorily right so like you know Aristotle thought the things that were of the earth element fell to the ground right and that's why that's why they fell to the ground because they had an affinity to the ground because they were of the element earth and inversely for flame right and so so for the ancients fire was a the element fire was like the underlying kind of commonality and explanation for what we would regard today is very diverse phenomena all right like so bioluminescence and little insects like fireflies that was fired right rapid oxidation of wood material in like a campfire was also fire but the Sun also right which we know is in fire I mean it's it's nuclear plasma right that was also fire so the progress of science has shown us that well this concept that we were using doesn't actually have any explanatory grip-it names things that are heterogeneous in their mechanistic basis and so similarly one might argue will life is one of these concepts like it it seems to us at a kind of everyday macro level that there's a distinction between life and life but is there I mean III don't know and III sort of started to think that these like working on these concepts per se is kind of moved and so like III don't know if I want to give a blanket definition of something like cognition yeah because I don't know if there's such a natural kind to begin with right I don't know yeah yeah yeah well a lot of these categories of concepts I like we the way I think about it is there's just patterns that we interpret right and there's and we just give a label to some patterns and like you know all dogs have a certain there's there's a certain pattern that they all have in common but there's a lot of like just crap there's a lot of noise it like doesn't actually correspond to what actually you know like a chihuahua might seem more like a rat than a you know a greyhound for instance but there's still dogs so like I think of these concepts is just like approximations that are like good enough to enable us to work with them right like you can have you painted me in a corner right and you put a gun to my head and said define cognition I would say something like it's information processing in the service of maintaining phenotypic integrity right in the service of keeping me alive I mean and so that's another angle into the free energy principle by the way yeah yeah I want to come back to it I just thought I would kind of like go through these yeah because there's a lot to it right as if we're just talking about entities or organisms like maintaining themselves or like trying to minimize surprise does the discussion of like how they actually do that what they are like how I mean if we're talking about extended cognition of whatever it was whatever it was like that's if you're really extended out there then what is actually being maintained so I don't know where I was going with that but um if we were to bring it back back to the free energy principle and like in terms of just minimization of surprise with these concepts and discuss though for free yeah sorry I'm sorry that's right you know it's a lot to take in and it and like you know for that for better or for worse like it all feeds into this yeah so it may be good to put all the so okay so effective extended and embedded the two remaining is our I think like the most exciting personally the embodied and inactive so the embodied mind is this idea that cognition isn't just in the head like cognition doesn't just isn't just accidentally something that's implemented in fleshy like wetware systems like cognition like is premise on the existence of an embodied organism and the basically the embodied mind says something like well a lot of well you need to appeal to the physical living body to explain a lot of what cognition is right so you can at least take this and I think three different ways one one would say well the physical body is Co constitutive of cognition so I mean you may have seen these really depressing studies that were conducted on judges um a few years ago like where they after lunch or something they just miss her so much basically what they do is they plot it so they it's a survey it's study of judges and their acquittal behavior so like basically they they will they plot your odds of being charged or acquitted against time of day and so it seems that at the start of the day you have a basically eighty percent chance to be acquitted and as you approach lunchtime it drops to zero percent and then there's a bump back up to eighty percent after lunch so it it seems so that the way that the researchers interpreted this is that it has to do with blood sugar levels like when you have a lot of blood sugar available you're ready to consider things like more deeply and really and then when you're getting hungry there's just no more glucose to do the activity so you don't so if someone wanted to say well you know what cognition is is just like rational purely intellectual activity well I mean you know this is one of the this is one of the demonstrations if that isn't the case I mean it seems that the body essentially participates in the ignition another example of this kind of physical embodiment and its relevance is like phototaxis and crickets is a very very widely discussed example in the literature so phototaxis is a behavior that crickets the Crick is basically orient themselves towards mates by exploiting the geometry of their bodies in picking up sound so basically to initiate approach of a mate a cricket uses two neurons just two neurons and it basically computes which of the neurons is stimulated first and it moves towards the that one because it basically its ears are organized in such a way that the sound will hit one here before the other and there so the the actual information processing that the cricket accomplishes is to orient itself towards mates exploits the physical geometry of its body like rather than having to engage in a more complex computation the the actual layout yeah human walking is very similar I mean if you pay attention to to this too much you might hurt yourself but like basically walking the human gait is falling forward and catching yourself with the next leg so it just it is not the case as as some people would say optimal control theory in motor control study it's just not the case that you have to compute the explicit movement of every part of your limb like you're just falling and the only thing you need to compute is a limb trajectory such that you can catch yourself as you fall so examples of this abound it just doesn't seem be the case the like you can think of cognition in a kind of disembodied way so that's the street just to be explicit disembodied being like brain-in-a-vat just all the sensory of what comes in when I say if any brain appear in the extreme view but also I mean you know a lot of the cognitive science like Fodor style modularity of the mind you know like all these views that say that it's the sandwich model that's being criticized to be right that like you know you passively intake some stimulus that you you construct an internal model and the and so it's it's this internal model thing the the embodied mind people are kind of pushing against it's like I don't need an explicit internal model of how every one of my joints moves in space right I just don't I don't need to compute that my body is such that the only thing that I need to compute is where my leg lines to to absorb like to keep me from falling right but so that that's one way that you can understand the embodied mind another way that you can understand it is to say that oh this is a more first-person point of view is that like the the my first person understanding of the world borrows from my embodied disposition so think of metaphors like descent into hell or rise to glory or something it has to do with the way that my body you know experiences itself in its live space all right so when you're sad you know like the entire geometry of your body is kind of like slumpy you know like downwards we associate downwards to disempowerment and that has to do with the way that we experience space as embodied beings from a first-person point of view so the last way you might understand it is in terms of like social embodiment so this is kind of hegelian it's this idea that like the literally the the layout of our worlds embodies something about our culture so you know like a I'm sure it's for having been I know it's the case in Sydney it's also the case in Montreal like you know our our streets meet at right angles we use things like traffic signals and traffic signalization red lights all these things that effectively embody norms so that I mean and this is where it kind of connects with the extended mind literature like there's another sense in which the mind is embodied it's physically realized in these artifacts we yeah the the embody not just knowledge but like you know ways of engaging the world that like support and ground our coordinated group practices yeah and so this this releases the free energy principle pretty directly I mean maybe maybe I should hold off too before just jumping into that you know was there a final ii am i yes yeah okay good is that like sort of just too many used to keep track of its inactive negative yeah inactive means the cognition is something that we do so i mean the idea there is to emphasize like the sensorimotor loop and to kind of undo the sandwich model so consider the following like what is a visual cicada right yeah yeah yeah so your point is like our eyes aren't static but they're constantly moving left like very very small amounts and that's necessary for vision yeah it's it's it's the mechanism of vision really it's like a lot of what your visual cortex is doing in interaction with the motor cortex is kind of deciding where to look next right so there's a sense in which that the classical sandwich model tells you action sorry perception is in the service of action right and it's it's it's linear it goes one direction writes with perception cognition and then action and what the inactive approach tells you is well actually action is also for perception it's really aloof you know like it seems like specifically superior as well because like you learn a lot more from acting than you do from just perceiving like precisely from from acting you get all the information maybe know little you get far more than you would if you're just perceiving um yeah I think that's really where the active inference stuff gets its teeth from yeah like active inference is ultimately a theory about how you sample the world so about how you engage in specific actions that have an epistemic value hmm and that help you design big you ate the world yeah more understandable accessible and empowering us stacks within it in a way that's hopefully beneficial for ourselves and does results in catastrophe precisely yeah and so the I would evolve of them I think like you you can tie all of these ease to the free energy stuff I mean we started with the effective yeah I mean there's a lot of work you know from increasingly from our group but mainly from other groups right now tying this as free energy stuff explicitly into mechanisms of effective processing so just a question on free energy why is it called free energy because my understanding of free energy is like energy that is within a system that can be used towards some end like yeah so the heuristic aliy the way you can think about it is that the variational free energy is a measure of how much wiggle room you have on your parameters still to get a better representational grip on whatever trying to represent so there's a direct analogy in that sense right if the thermodynamic free energy is the amount of energy left in the system to perform work the variational forget energy is the amount of energy left in the systems that keep doing like better representational work I mean it's it's worth mentioning that like they are the same quantity in a certain sense like the the the information theoretic variational free energy is just a generalization of the thermodynamic free energy okay yeah I mean that's where it's a little bit that's where it gets a little complicated and that's where like a background in physics is kind of useful it has to do with these these deep relations between probability and energy so you know this this goes back to like Boltzmann and stuff like that like like statistical mechanics so like if you think of a volume of matter right like the the more energy is in the system like the hotter it is like the the more possible configurations it could be in right so like if you think of a gas and gas like you know it's very hard to predict where any given molecule the gas will find itself in the volume it can move a lot if you think of a crystal very regular relations right so the the notion that we use in that kind of context is entropy right so entropy is a measure of spread essentially so if you think of a probability distribution right over a set of outcomes right so like I don't know to make it very simple I'm gonna I'm gonna pull a ball out of the Hat and the ball can be one of four colors and a red red green blue purple like if I if I don't know anything at all you know about about the balls like that it's so there are four kinds of balls and our our bag and I'm picking some out there are not just four balls though in the bag there like as many as I as I need so I want to know something about like the distribution of balls you know of ball colors well so if if all of them are equally probable like if in reality there's like 25% of each type right well the distribution is flat right like because there's 25% of each type so if you drew a histogram like it would be a flat distribution a flat categorical distribution right inversely if you know there was only one one red one blue and one green and 97 purple out of a hundred balls then you would have a very very peaked distribution right like it would peak over purple and then it would be very low over the other colors so the first kind of distribution is not informative right like it's you know it's equally probable everything is flat we say that the entropy is high in that kind of context because the probability distribution over the system is flat right and so there's a lot of spread inversely in the second distribution the entropy is low because the probability is basically all concentrated over one outcome so so it's very informative and so you get this kind of behavior in physical systems when you consider heat right so a very hot system is high entropy the the specific particles that make up a hot system could be in any number of configurations a cold system to the contrary has a narrower range of states in which it can be I mean you can see this literally as water turns into ice right like if the degrees of freedom of the parts are drastically diminished as they end up forming a solvent right so the so there are these kind of deep relations between the probability of finding the system in the given state and the energy in the state and so the free energy principle kind of builds on that it I like to say it kind of jokingly that it answers the question how can we be so cool meaning that like as living systems we were we're in the information theoretic sense were very cold systems like so the the the anthropologist Terence Deakin makes an interesting point he says you know contrary to the received dogma and philosophy the whole is much less than the sum of its parts consider an engine right for an engine to work each of the parts that make up the engine have to be constrained to behave in very specific ways right like you wouldn't want to start introducing degrees of freedom into the piston movement right because you would you would get some very serious damage done to your system so so like organized systems like engines and humans and bacteria like live in very restricted like subspaces of the possible space of configurations that it could be in like so you know in in theory you know my body temperature could be 50 degrees Celsius or 3 degrees Celsius but for some reason it maintains itself at 36 and a half degrees Celsius and you could say the same for a lot of the parameters that make up my my body and so the question is well how do you manage to stay there and that's where the free energy principle comes in so it's not like organisms like we're complex adaptive systems rather like life is well we are you know remember we are complex adaptive systems and because we are ordered in such a way we have a low entropy I'm just want to make sure that like communicating this correctly we have a low entropy because the probability of finding this arrangement of atoms in this state is incredibly low and that means we have low entropy right and what we do is in order for us to maintain this state across time based on the fact that the second law of thermodynamics states that entropy is increasing across time that we need to be taking energy in processing it in some way to ensure that we maintain this low entropy state across time lens so how does the free energy principle fit into here is it that's where we're getting right yes the the free energy principle tells you what so what keep a lot of people don't get about the the variational framework is that it all rests on a bit of math so it doesn't make sense to say like you know the is is is the free energy principle true or false it's like well it's the same status as calculus it's look like you know you don't you don't like disprove calculus like if there's a problem with calculus you derive a contradiction right or are you sure that this or that kind of bit of the formalism doesn't follow and the free energy principle is very much the same thing it like if we're really just drilling down and like just looking at it from the most technical point of view what it's basically telling you is what must necessarily be true of any system that exists if non-equilibrium steady-state so let me unpack that equilibrium is where is the end point right so the second law of thermodynamics is about equilibrium essentially it's telling you like we're systems were cut we're almost all physical systems are kind of converging towards those where like we just think it's a fun plate right like once out a hot plate it's hot there that's cold over time the home plate will be the same temperature it all it's not just the nature of boors a vacuum is that nature abhors a gradient right like nature things to be perfectly flat so again it's the entropy thing it's like if nature is trying to flatten all the probability distributions right so I I'm a big fan of ginger ale I like a nice cold fizzy ginger ale but much to my chagrin if I leave it on the counter for 30 minutes it'll become room temperature and not fizzy right so this kind of room temperature not fizzy is equilibrium right it's just like the the and you know if you pushed it to the extreme and you I left the thing I left the the you know the glass of ginger ale on the counter for two billion years well it will have completely dissolved and sublimated into the into the environment I mean like after two billion years like potentially like even the solid material of the glass would have started to like diffuse into the environment like through sublimation there might not even be a glass there left anymore so one thing to notice about systems in nature is that self-organization itself is almost always in the service of increasing entropy and returning to equilibrium so that that fact is underappreciated I think in in the study of self organized complex adaptive systems like so think of a lightning bolt alright so lightning bolt arises because you have a charge gradient right so like you know it because of like the movement of particles and whatnot like you get more electrons stuck in one place than in another and the in striking the lightning bolt corrects the balance right sorry corrects the the imbalance and it brings the system closer to equilibrium so the same is true of a tornado right so you have a temperature gradient in the tornado yeah typically have very cold and very hot air that kind of meet in a way that they're stacked and then like the tornado it looks like it's increasing order locally and it does locally but but the cost of it is the increased disorder globally and the same is true of the lightning bolt the lightning bolt self-organizes around a charge gradient and in striking it consumes the gradient around which it was self-organized yes so this is true of almost every self-organizing system they increased entropy and they consumed energy to stay to sell self-organized so if you think of the star for example you say increase the entropy do you mean locally or do you mean global good yeah yeah because that's the important distinction right I was outside and confused because I'd like to know well stuff organized systems want to say self-organizing they process or using they use energy to put it crudely think of how much think think of how much matter you have turned into poop over the last year yeah I mean just just dirt like think of how much literally the raw amount of matter that you have disorganized in order to maintain your own integrity this is this is the rule I mean yeah is the rule right it's like yeah exactly a lot of energy right like it I mean all of the energy on planet earth basically used by living organisms except them like these weird sub marine ecosystems come from the Sun but I think the the figure is something like it takes 10 times more energy to create a star than a star will output in its whole life in terms of like the the actual energy that it takes to create a star right like everything kind of yeah so even a star is is a is a machine that primarily serves to increase the entropy of the universe so having said all this you know living systems like bunny rabbits and tigers and and you know you and I don't self organize to equilibrium we self-organized to this other thing that's known as a non-equilibrium steady-state so the difference is kind of intuitive let's take a simple example my body temperature is 36 and 1/2 degrees my core body temperature that is and so that's a steady-state like my body set that as like a point and my homeostasis makes sure that I maintain myself at that point so I mean this is a contrary - like I was saying basically all other kinds of self work in my systems the don't self-organized to this steady state but rather self-organized to equilibrium for for organisms equilibrium is death right if I became if I if my core body temperature kind of diffused to room temperature I would die like quite literally so the free energy principle basically tells you what must be true of an organism if it exists at non equilibrium steady state and so it's it's literally starting from information theory and dynamical systems theory and then like formulating ok so like what does the system look like when it exists at non equilibrium steady state in these regimes and what must be true of it for it to continue existing in this in the state in a nutshell that's what the free energy principle does so how do we begin to there's a few there's a few ways I want to go over here one is just describing or explaining how we'll be sort of measuring these things or like is this found in all I guess this is when you mentioned it's kind of like a tool like calculus it's it's not a thing that's observed it's a way of observing or way of describing these things right and this is just something that was a characteristic of life so this is where it gets a little subtle it's so that having said everything about the non-equilibrium steady-state okay so I didn't tell you what the free energy principle does I just kind of set up the problem that it answers right so to recap organisms existed non-equilibrium steady-state right so now we know what that means right so they don't they don't self-organized to equilibrium which would just you know it's it's my flat and room temperature you know Jenny right all right III self-organized to my phenotype I self-organized to the set of states that I occupy most of the time right that make me the kind of creature that I am right so if I'm a fish I self-organized to a set of states that allow me to exist as a fish right so I'll grow gills right and fins and so on and you know analogous things can be said for you know self-organization as a human so given that I exist a non equilibrium steady-state the free energy principle tells me what must be true of me you know provided that I continue existing at non equilibrium steady state so in a nutshell that the free energy principle says is that if I exist at non equilibrium steady state it will look as if my behavior is being generated by a statistical model of my environment so that's the crucial point is that it's are you familiar with Dennett's Daniel Dennett's like the intentional stance no okay so it's a position in philosophy the basically says well what why why is it that we we are able to treat each other as intentional systems ayiiia systems that have aboutness right that are able to respond adaptively and intelligently to the environment so then it says well essentially it's that we we we adopt the intentional stance it's not that there's something like intentionality in systems it's that we we for certain purposes adopt towards others a certain perspective that describes to them beliefs and desires and so on and for our survival and and insofar as our capacity to explain the behaviors of others is involved its adaptive and efficient to do so right so this is the so called intentional stance so the free energy principle has a similar flavor it says like it looks as if organisms the behavior of organisms is being generated by a statistical model that the organism is so yeah and the statistical model like I just I really like the idea of like I think the most intuitive way for me to think about it is the minimization of surprise which you could think of as this if we are a statistical model as in that we are like as you said it's not just our brains but our entire being is the model then I like this notion of minimizing seeing that as a way of minimizing surprised because surprise is not always good right especially for life surprise could be a lion or it could be a disease it could be anything it could be I mean in terms of if you're looking at it like just very technically the way that it's set up is that like surprising the states are not good for the most part and we're talking about internal states but no oh yeah yeah okay so I mean it gets a little subtle because humans are also averse to boredom for example right so yeah it's just about making sure that you never are surprised right it's just that there there are certain kinds of observations about which you don't want to be surprised yeah right so like I currently observed that there's a lot of breathable air around me right I'm breathing very freely right now similarly like my observations are consistent with my belief that my core body temperature is thirty six and a half degrees that I still have four limbs right like that so that there are there are some beliefs about the typical things that I should be observing that like you know challenging those beliefs is very bad for my survival right like so if suddenly the first belief that I mentioned right there's a lot of breathable air was suddenly like put into question by by sensing evidence counter to it I would probably act in a way to make sure that yeah different was gonna happen because I I need air so that there's a sense in which like the free energy principle builds in like a good part of the the normativity involved in existing as a as a prior preference over certain kinds of data or like observable outcomes like that I fir - generate certain kinds of data that doesn't mean though that like I I want to eliminate surprise entirely know what that makes a lot of I mean because you for one you can't and two surprises informative right and as you go up scale so this is where I'm very excited about what the reason I wanted to speak in I think it applies to some of the work that you've done you know about how this is found across scales and yes that every time but given that we are embedded in our own actions change the environment that as we increase as the scale increases there is always more to know and we can always be surprised and now then we're at this as planned now we are a planetary species we have a whole host of things to be surprised by now that we there's a sense in which like the so this notion of prediction error it effectively replaces the the classical notion of signal so consider the following like a one-to-one map like a 1:1 scale Knapp would be completely useless right like I mean imagine like you know you're visiting New York for the first time like a one-to-one scale map like you know it would be it would be like bigger than a building right like the actual amount of paper so like in order to be useful at all a model has to be simpler than the the data space that it's modeling right you know like so there's a sense in which like just by construction for a model to be useful it will have to generate error and so in the kind of this is slightly broader than active inference and the free energy principle it's the Bayesian brain with slightly older you know approaches developed like in the 90s and 2000's so like from that point of view signal like the signal that the organism perceives is a prediction error like because basically the idea is like if if the if my sensory observations match my expectations there's no sense in processing it it's just it's what I expected whatever there's no sense in a noble izing my metabolic resources to address a percept that I had perfectly predicted and that you know if I'm hooked up in the right way like you know addicted perception just kind of roll off you know yeah yeah I'm needing to engage them yeah so that there's a sense in which like all signal is error it's efficient that way because yeah yeah yeah yeah and that's that's what I'm Lisa Feldman Barrett was that's what she I don't think she discovered but that's what she spoke about a lot I kind of forgot about that part of it that's um that's really interesting it underlies a lot of the like motivation for the approach right so we're saying like an organism embodies a statistical model of its environment meaning that like there's a kind of average value that it expects to encounter for different classes of stimuli right and the organism will only be prompted to process like a stimulus when there's a deviation from what it expected yeah and so there's an extra twist that makes it especially interesting and that makes it robust to some objections it's not as a darkroom problem I think it's an on problem honestly but so the Darkman problem says okay well if there's this surprise minimization thing going on why isn't it the case that I just find a dark room and stay there because that would be a great way to avoid all prediction error right if I predict that I'm in a dark room and I'm in a dark room there's no prediction error and so why don't I just do that two reasons I think kind of spring to mind immediately the first is that you can have an X I mean so we're embodied right to bring it back to the embodied mind like my body is a statistical model like it's a I embody expectations about you know my environment so for example I have expectations about my blood sugar level right and so staying in a dark room with no food is gonna you know frustrate some of my expectations about my blood sugar level such that at some point I'm gonna you know move out there are more direct ways to gerrymander it a bit and I can see why you might find those like less principal you could for example just stipulate that you have an aversion to you know encountering no surprise for a while there's no reason why you couldn't do that like just to bake it into I mean but there's a more principled way to do the same so this free energy quantity that we've been talking about what you can do is kind of think about how it evolves in the future so you can think like for every possible course of action that I can take how much free energy is associated with that with that course of action so when you have this future a future-oriented or this kind of this extended construction of the free energy it's called the expected free energy it turns out that for some using these nice mathematical tricks you can decompose that into basically a pragmatic component which is basically how close do I get to my preferred observations and an epistemic component which is irrespective of how close I get to my preferred observations am i gaining any information by making this observation so like I don't know it's it's 3:00 in the morning you come back from the pub you're hungry right you stumble into the kitchen right but it's 3:00 in the morning so it's dark the first thing you do is open the light well I mean from a kind of classical kind of expected utility theory perspective that's a bit puzzling because you don't get any immediate gratification from opening the light you know it's not like it's not like you're getting like you know like sexual gratification or like you know food stuff immediately from opening the light the old but we do though right like you know we open the light and the reason we do is that like there's information gained from doing that so so an action basically has an pragmatics component the the pragmatic affordance or a value of a of a policy and the epistemic component and the value or policy so you another way to respond to this dark room objection is to say well look like the organism also acts to basically increase its epistemic grip on its world so there's a point at which there's nothing more to gain from just like staying in the darkroom and so you'll probably you know here's something outside the room and want to disambiguate what that is you know yeah I'm curious about what the implications are or what we know about this and social species where species have to coexist and collaborate you know they can't exist in isolation they need these organisms need to interact and cooperate in order to exist and I'm thinking of you know bees and ants but also you know humans um what does I worked a lot on this like that's sort of one of my the main thrust of my research is extending this active inference business to a social and cultural behavior and I guess the main idea there is that these these models that I was talking about earlier these statistical models so they're called generative models because so technically these models there they are the joint probability distribution over systemic States and external States so systemic being the organism yeah exactly it's internal States it's active States its sensory states and some some set of external states yeah yeah so wait sorry I lost my I lost my idea there yeah I think I've been there once or twice this conversation - we're talking about social species I'm sorry yes so they're the so these bottles I was just saying this because they're called generative because because of the setup that I was describing earlier right you can use these models to generate the kind of data that you would expect if the model was the case right so like having a generative model kind of sets you up with I don't want it like I don't necessarily want to say normative but the kind of it sets you up with like the a a grip on the kind of typical observations that you would make given the kind of creature that you are right so that's that's literally what what it does it's a it's the joint probability distribution over some external states right and some organismic states so the way that we do cultural and social dynamics under this approach is to say okay what happens if several agents share the same model so again like the model essentially harnesses your expectations about the way states of the world which could be social states hookup to the kind of typical observations that you make right so one way to talk about sharing the social world is in terms of sharing a generative model ie sharing a kind of statistical model about how the observations that I make typically a pond two states of the world and if if we were to just translate into everyday language like people seeing the same phenomena in a set in the same way exactly sharing the same expectations yeah and prior beliefs about the world and about how my sensory phenomenology relates to aspects of the world yeah nutshell so I mean there's some pretty compelling simulation work done on it's by Chris frith and Carl Fuerst and mainly although it's been extended on basically little agents it's like birdsong simulations so you have a little agents that sing a song and they're able to coronate like little simulated birds they're essentially able to coordinate their singing behavior and recognize when they're singing versus when the other bird is singing by sharing the same generative model yeah so I mean the idea is sort of that if we share the same expectations about the way that states of the world map on to our observations and the way that states of the world involved then you know there's a sense in which each of our coordinated each of our individual behaviors end up coordinated because we expect the same outcomes right yeah so so that's how we and Dewey so it's like I think about it like this if we're all I don't know drivers are something wrong people and we all have a map if the map is the same and we're kind of working with other people who kind of go on the same roads or in some if we're all following the same map we'll end up going to the same places if there's like some sort of directionality or you know X is the spot I know that I've got a compass is the map I will go in that direction at everyone else will but it's exactly not the same and people might come that's the and yeah so there's some cool results this is Eddie bulik who's currently at UCL she's done she's done a lot of hyper scanning are you familiar with this hyper scanning or socials guess it's a multi brain neuroimaging it's like the new thing right now so like rather than just scan one brain yeah like in real-time you hook up several people to the fMRI or EEG or whatever your modality is and you you can you can measure brain to brain information transfer yeah it's it's a cool it's a cool do a lot of what we're doing now is gonna be combining like this hyper scanning set up with active inference malls that use multiple agents so that we can I mean are the participants doing things like like I first think of like video games sports choir jazz bands it depends I mean like there's been hyper scanning done on people the cool thing about some of this hyper scanning stuff using more portable technologies that you can have ecologically realistic settings so that they've used EEG which are these like caps that pick up scalp electrical activity in operas for example to get like people's real-time impressions of like seeing a show together in this case it's it has more to do I mean these are techniques they're used in several different studies I mentioned Etta's work because she has some cool data where like it seems that wrote I mean this is one of the things that I have to check with her if we can put it on the internet yet so we might have to cut this out but this may just be for your own benefit but like she has this cool work that shows that basically romantic partners don't don't they don't attuned to each other anymore they don't have to like that they're not actively trying to adjust to the other person anymore because they have they have such an accurate model of the other person so there's like a sweet spot there's like those predictions if your models are not similar enough like if you don't speak the language and you're not from the same culture there's not enough commonality to start adjusting but if you're if you're too similar there's no need to adjust yeah yeah yeah so there's like an inverse you curve kind of thing going we're like on the one end like there's perfect synchrony because you have the same model so this it's like these romantic couples that don't really need to check with each other anymore because they just know they know what their partner thinks right it's like oh people they just don't talk to each other because they're just telecommunication exactly yeah yeah so so like the the kind of overall framework you know kind of gives you this inverted you where like you know the yeah it's all about the extent to which you share a generative model and so yeah I think that that kind of well illustrates like this the kind of generalizability of the approach yeah yeah I be we I mean this is a bit sad I guess but be funny to see if it predicted divorce you know like incoherent models just like I'm sure that I'm sure they would right because that's how a lot of divorces arise right people acting in one way that doesn't you know here with what their partner wants and over time yeah I mean so the these these techniques I think are being scaled up more like you know interesting you know socially rich human behaviors yeah and yeah so it has to do with the full pipeline I mean so I've been I've been talking about like the the side of this that I'm like coming from Moore which is like the the kind of computational modeling side but I mean to to return to the if you don't mind me sharing my screen for a second again place place I'll of course describe what's going on yeah this is what I wanted so the the whole the full pipeline that we're considering is this kind of thing where like what I've been talking long mainly so what we're seeing now for those of you listening is three circles so you have two circles on the side that are pointing towards a middle circle the middle circle is data and the two side circles are the modeling bit that I've been talking about a lot in the experimental side of things that I I alluded to a bit when we were discussing this fMRI stuff right so like and this relates to our earlier discussion of like you know is this an explanation really like the full stent the full pipeline is this thing here it's it's a combination of experimental and modeling techniques that are kind of joined at the data so basically you you start with either end so suppose you start with an experimental setup like we were just discussing your your which you have is like people performing a psychological task and you're getting either you know neurological or behavioral data are both from them right so that's this side of the equation here right so you have an experimental setup that generates some data and then what you can do is write down a generative model of the kind that I just described or rather what you what you end up doing is writing families of generative model the kind that I just described that each represent a hypothesis about the causal structure of what generated this process that actually generated my data so like it so in this in this instance like each of these models is is a model of one possible hypothesis about you know the the causal factors involved in the experimental setup the generated the data in the first place right so you construct these models you score them using the variational free-energy to find the most probable one right and so that that model that you end up selecting encodes a hypothesis about the structure of the process involved in the experimental setup so now what you can do is kind of bootstrap your thing right like you can use the fitted model right to make better predictions about what kind of things you would have to test experimentally so then you can refine your experimental setup and generate better data and then using this better data you can construct new models that are more refined to make pointed hypotheses about the structure of the process that you're investigating and so you get this cool kind of bootstrapping self REITs like virtual racing feedback ions yeah that's really that's an underappreciated point like you know III I see these kind of partial criticisms of the active inference framework that go will you know like surely these computational models aren't a full story you know like you're you're not really talking about the mechanistic implementation which is fair it's but the thing to keep in mind is that the full framework necessarily includes this neuroimaging aspect that makes very pointed hypotheses about brain structure for example yeah and in fact if you think about it the origins of the approach you know are in fMRI so like the the behavioral modeling thing that's kind of become front and center that's gained so much attention is just one part of this whole process I mean so as a philosopher the thing that gets me really excited recently is this thing computational nosology so this is where we're moving towards a kind of meta science or a meta Bayesian meta science so I mean to rehearse the argument you can have if you apply this framework to some set of experimental data like a bolt signal then what you're doing effectively is writing a model of the underlying neurobiological activity that might have caused your data right we've discussed this active inference is takes the same kind of approach and says okay here the data are the sensory signals that I generate in engaging with the environment and the model is a model of me acting in the world okay so what what happens when the data that we want to explain is the result of the scientific activity itself so for example what happens if the data that I want to explain is a psychiatric diagnosis right what I want to explain is the diagnosis of you know the actual diagnostic act of you know saying telling a patient you have schizophrenia well then the model is becomes a model of the diagnostic process and then lo and behold we are part of that model as clinicians and experimenters right like the the generative models that we use in this sense of the active inference sense right the the models that I bring to bear as a clinician to make sense of the clinical phenomenon and presented right and also these models like the models of how like the the hemodynamics and so on like real fever exactly all of that is oh in this kind of meta Bayesian approach where like the the full model of the diagnostic process includes the models that I use this as a theoretician and the the clinical models that are used by like a psychiatrist so this is where we're going like so it seems to me it's just like the scope of what is actually important is far broader it not just the actual experiments of the models themselves but it's those who are actually well running them and actually coming up with it yeah so wet is where's the boundary what is it I mean so remember the the free energy principle is a theory of a non equilibrium steady state so if if a system exists at non equilibrium steady state it can be explained by the free energy principle this isn't just a hypothesis by virtue of what it means to exist at non equilibrium steady state for principled reasons you can appeal to the active inference framework a very cool stuff to do it so the extent that science is just another you know large-scale human activity with its own not equal to Rijo steady states then it can also be model yeah well this kind of brings me to the point about social systems and and all that and I guess taking this and applying it to our politics and our societies and I mean we all have is aligned I'm working right now with my PhD student on the constitution of epistemic communities so like it sounds very abstract in one sense but like so think of like you know you seen these maps of like the the us like electoral politics and everything where you have like these little islands of blue and these seas of red right yeah so close they are of the perspective that we're advancing here I mean it may be that like people are literally living in different worlds in the sense that like they are no longer the same kind of creature they don't they don't have the same generative model like just the we we've effectively you know the the course of politics and for example the United States but I think increasingly elsewhere especially I mean like the information environments in which we're living in I'm just so like because we can select what we you know through Twitter Facebook and how these social media environments we can construct our own information and they in turn construct us and what I think is yeah and that is a beautiful point and that this is one of the things that we've really drilled down on on the active in the active inference family of things I mean you in the discussion before we started you mentioned axial constant who's a really close friend and collaborator an axle is responsible for also with the yellow Brandenburg but Axel is like one of the people who's really championed this for the idea that well so the active Infernus formulation is symmetric you know I was telling you about the this predictability stuff earlier that like you can predict stuff about the environment given my phenotypic state well not symmetrical okay like you can predict something about the kind of things that live in an environment by looking at the physical structure of the environment like the relations of prediction are kind of circular in that way and certain creatures like ants and beavers and humans have evolved such that like we we extend our statistical models into the world like we we we ought meant the physical world with but by allowing it to encode information that we can use to guide our behavior so think of like you know the fact that in most big cities streets meet at right angles isn't accidental right it's a feature that we design right into the physical layout of our cities to to improve our predictive grip on them essentially like to make it to make our coping with them smoother same goes for like you know traffic signals and stuff like that so I mean there's a sense in which this story opens on to something like and this is what we're exploring in in our current work like the this kind of opens on to a story of epistemic communities and like epistemic Miche creation right like we we we design environments in order to better like to get a better predictive grip on our environments and that might explain in part like the kind of fragmentation that you see right like that we're literally creating two different kinds of worlds that are embodying two different kinds of well outlooks or we would put it in terms of different generative models yeah yeah yeah a direction that I'm very excited to pursue actually I think you know what there's a lot to say there sure and when you consider that so right I haven't made a public yet but it's um it's a I guess it's two articles one is just on the Internet and control and why how it's a human rights issue and one is that the Internet is just necessary for securing human rights and secondly we do not like our access to like it is it is so vital today but our access to it is actually well it's under threat at all times we may not know it but like it could be taken away from us at any point and in some places it has been I mean China is the greatest example of government censorship but there's like dozens of countries have stopped social media or like interfere with social media and messaging apps and when the state or went you know privately held interests can effectively dictate what is in one's information environment they can do a whole range of things from manufacturing consent to well you know just changing how people think and and and how they acts in the world and I think it's a it's a huge huge problem and one that like I don't really know what the solution is because if you know if you take the liberal idea that you know we should be free to go out and do is do whatever he wants within reason within certain bounds but you know I should be able to expose myself to whatever information I feel like as long as it's you know not hurting anyone or anything if that's actually if we agree with that then we can't actually impose these informational constraint saying that you know it's all well and good for each for you to go ahead and do that however we need you to be we need our generative models to be somewhat coherent or we will conflict yo or things just break down yeah like in a really deep way yeah I totally agree I mean you know what a huge issue is uh whose principle is it that it takes it takes an order of magnitude more energy to dispel than it does to create it right someone's rule I forget I forget what whose rule it is but I mean yeah the that's a real issue I think you know the I mean I I I'm definitely not a liberal I'm not a conservative either so I think you can kind of situate me you know pretty clearly uh yeah no I I don't I I'm sort of grappling with the same kind of issues that you are here like I yeah I also value freedom of speech and freedom of expression but on the flip side like there is something like there's like an ecology of memes you know like the there there there's only so much like physical space that ideas can occupy right so like that I mean you know minds are like in on the net oh yeah on the net and like physically in space yeah as well they're technically the same right like ah yeah I mean that is an embodiment yeah exactly it isn't Bartek yeah so I mean I'm sensitive to that I I'm very worried about like the creation of like these fact resistant you know bubbles uh but you know like the there's a sense in which like I think the the current Information Age exposes a lot of our kind of innate shipping inherited like evolutionarily old psychology right you know like the what gets called virtue signaling oh yeah I think I think happen is both on the left and on the right the show official I think ultimately the problem is that like we we have forgotten how to engage with ideas and what takes the place of engagement with ideas is signaling to your in-group whoever your in-group is your values are consonant with theirs in my last conversation we kind of touched on this and one of the observations that the guest Matthew Bukowski made which is you know very very passionate and and true is is that social media kind of it rewards signaling but not action yeah like we're all engaged you know we're incentivized to just signal great things right things about our lives or our ideas or whatever and that signaling is enough it doesn't really matter if we're going out and doing stuff and we need to move from signaling to action especially in the filter to your Facebook I stand for you know breast cancer victims like oh you've just saved it you've just your cell with it and you know I think we're all guilty of this I am myself absolutely and that's what I'm saying like I think it's a it's an issue everywhere it's it's an issue you hear about it a lot because like right-wing pundits accused like left-wing activists of virtue signaling a lot but I mean III think you can see the same essential cognitive strategy redeployed everywhere essentially so I mean that that's that's something that's sort of unfortunately predicted by the kind of theory that we were working with oh that's brutal yeah well because like what you're what we're saying ultimately is the work action is in the surface is in the service of generating evidence for our beliefs essentially right so like things that things that conform to our broader yeah so like it it takes a lot more cognitive energy to like confront you know observations and data that don't provide evidence for your belief and in actually her I mean in literals yeah yeah and I think it's like I think of this as like we are constructed by of we are information right like who we are these are the ideas like the person I think I am is a collection of these ideas and I've kind of brought together and right I identify with and if I find something that I disagrees with them I am actually being attacked in some in some way right and I think that's where that pain comes from because it's like there's some oh yeah I made that is was a lot more complicated because I mean you know left and right what you're seeing today is a rise of identity centered politics I don't want to repeat like you know kind of dumb right-wing tropes about identity politics yeah yeah it is true but like a lot of the politics that we see today on the left and on the right concern questions essentially relating to identity right yep so you know it's not virtual it's not first as we spoke about earlier um like that you can basically categorize something an infinite number of ways some categories are just kind of a bit better at like at whatever you're trying to do so on the one hand I don't want to compromise like I you know activism based on legitimate you know like you're suffering of a particular group in house right yeah the show of a showing it like you know denunciations of you know forms of oppression that are rooted and like I don't know in the u.s. like you know slavery yeah elsewhere like and you know histories of colonialism and everything I mean you know as an Australian in the Canadian we should know a thing or two about like colonialism right you know we come from countries where like you know it's some of the only like actually quote-unquote successful genocides have been carried out right oceans of blood yeah exactly where people have been systematically exterminated like and where we're like that was carried to completion I mean in there are places in Canada where that was carried out against four stations people and I mean you know the this is the actual genocide of the conquering like all that and you know the same thing with in Tasmania or the yeah so I mean I don't want to minimize those legitimate airings of you know like or but but on the flip side like when when you make your identity into your thesis it becomes very dangerous to disagree or to argue about that you know so like if your main thing is all if your main thesis is like us something about you know American identity real American patriot or whatever then it seems to me that like you know disagreement will quickly collapse into like yeah you're attacking my identity yeah like my deeply held sense of Who I am so I mean III also think it's probably not the most productive way to the frame discussion on the flip side like I haven't quite squared the circle of how you how you recognize that danger on the one hand and also recognize the legitimacy of identity based you know like legitimate grievances and being there a yes it's a tough one I think one way that kind of it doesn't solve it it doesn't definitely doesn't solve the problem but I think a lot of this discussion of identity is rooted in inequality problems of equality and I think there's too much of a focus on equality in terms of income and wealth even though it's a great proxy for well-being and all that it's just not good enough the quality of opportunity is something that we should really be focusing on more and as I said you know wealth is a great proxy for that but if you can't read it doesn't matter how much money you have right um you don't have as acts you don't have as much access to the world as someone who is literate or if you have shelter or if you're adequately nourished you have access to the Internet so by focusing on I really like the idea of focusing on equality of opportunity and seeing like the importance of increasing the landscape of opportunity to people because if you think it like freedom is in binary right freedom exists in degrees and if if we want if one of the aims of society is to ensure that our citizens are flourishing one way of ensuring that that happens is like you make them as free as possible like an increased landscape of opportunity as much as possible so they can go and choose to do things it's in to it in order to do that you need to ensure that they have their basic needs taken care of sir you know shelter nutrition the ability to read perhaps all these things increase the landscape of freedom and then the individual they know their map of themselves is the best right there's nothing better than their internal map of themselves so they will choose to live a life that maximizes for their own flourishing so if we wish to ensure that our populations are flourishing we sort of should kind of increase the degrees of freedom that I made available to them and let them navigate an unfortunate thing about neoliberalism is that you know in in in the current kind of set up what freedom means is like the freedom to choose between 36 different flavors of ice cream yeah yeah it's I think it's just a bit of it I don't think when we had these conversations wildly enough like what is the purpose of society know what is government working towards like what does a good life mean I mean I think you know we might have that conversation around a campfire with a joint you know but it's like so we don't hear like our politicians actually having these conversations all I hear is this retro jobs yeah I totally agree I mean so I've been setting up a an international Neuroscience network over the last year or so and we're really trying to foreground ethical reflection because currently I mean you know this whole ethics review board business is basically a box ticking exercise right we're like you know in science the way that it basically works as well you know when when they were like figuring out like these you know the wording of these Accords right like the regulate research on humans then they involved philosophers and everything but like it's sort of like they settled everything you know wet like at that point and from now we're not really thinking about like the ethical implications of our research we're just making sure that no one's gonna sue us yeah which is what the IRB is do so we've effectively evacuated ethical discussion from like the practice of science and so yeah ethics usually means in science like you got a board to review and it's pretty safe from a legal point of view like there's not gonna be any like legal blowback yeah well that's a very very thin conception of like ethical reflection about the scientific process I mean I couldn't agree more with you like the these things have really been sidelined like much to everyone's detriment I think yeah yeah it's I think we're gonna see a resurgence in well ethics ethical discussions ethical thinking I mean we were I think that we already have it's just like done through the guise of environmentalism and you know like fighting for the biosphere and you know humanitarianism so I take that back ethics is still there we just don't talk about as ethics for like we just appeal to you know human nature but I'm really excited to and this is one of the reasons we're I'm spoke to you what I'm where I wanted to speak to you and I think will guide a lot of the conversations that I'll have over the next future over the coming the coming weeks and months and years but it's just like what are the physical characteristics of things that are like what is the material basis for that which is good you know like if you if you take damn I mean you know well there's a question like yeah well I think like it's I think we have like some of a somewhat of an idea like just intuitively like systems that are complex like there seems to be a correlation between the complexity of the system however you wish to measure it and its ethical value right so I'm excellent I have an ongoing disagreement about this the the disagreement is basically axel thinks that morality itself can be absorbed under the free energy principle and I don't necessarily disagree but I want to push back against that as much as I can so for the simple the following reason like I think that like you know with the free energy principle etc will will tell you is where the system is converging how it's converging like what it is converging towards essentially right like so - no that doesn't mean that it's a good idea that things are converging in that direction though yeah see that's like this objective like there is this objective morality that states that that thing that is that you were converging towards is good right it's like separate but I see morality is like it's actually emergent as well right like you let me and maybe perspectival so if let's take this account seriously it sort of implies that morality is embodied in a shared generative model right that like has effects in terms of like you know allowable cultural practices allowable ways of like living in the shared environment but the implication of that is that like you know the kind of moral standards are ultimately relative to you know the shared generative model so that like what do you do then when there are groups of people who come into contact that don't share a generative model like I like I I find it difficult to because the scientist in need doesn't see like the the possibility of there being like some sort of external way to arbitrate between the sets of so it's irreducibly complex right and but we cannot put like it's it's too chaotic we can't predict any of this it's something we can just plug things into a formula and then say oh look this is the more ethical decision it's just like exactly impossible precisely I mean I you can only find the most likely decision no I think I think that's where these heuristics come in right like you know thou shalt not kill like these 10 commandments are like they're kind of just like guide rails for Humanity because a lot of it's really hard like we I think we've been trying to like figure out what we're like what is good and I think you know like these like the the the rule based stuff is just kind of like oh we've got a bit of an idea like 9 times out of 10 these rules are right the question of like I think there is like some material I think this can be described in terms of energy and information I'm just not too sure how but I think it's like if you were to take a word that's that's what Axl ends up saying I think I don't want to put words into his mouth but yeah the ends up saying something like well you know it is about constructing the right generative model so if if we can if we can just like get a picture of what the adequate moral system would look like and all share this generative model then you know we would end up converging to something like you know a socially acceptable situation so I guess this brings me to write the last topic I want to count before we wrap up and we're kind of speaking about it before but one of the implications so like I guess logical evolutions of this just applying of the free energy principle applying it to humanity is just like they need for like we are a social species we are engaging at scales that are you know increasing as as the years progress if we are trying to minimize surprise for you know to put it put it in those terms and one things one of the things that we really really need to do is integrate the sciences into our politics into our decision-making because like it is it is truly a matter of survival and we may be hit with some things that I mean they may not be too surprising to some but you know coded the array of challenges that we're facing with which actually threaten our not only our survival but like the survival of you know lots of life on on the planet like it makes what the free energy principle says to me is that we actually need to take these epistemic tools like you know science like these frameworks these ways of being in the world and extracting information and putting it to use and actually making them a part of our as our culture in a way that's far more like far more embedded and far more well I guess useful is sort of an ethical imperative the falls out of it I think with the free energy principle would tell you is that well if you don't do this you're no longer gonna exist at non-equilibrium steady-state right it's just that but that is where we're going collapse that's where we're going so yeah I mean the the the issue is it again has to do with generative moles it's like the generative models typically have a temporal depth built into them so basically like how many times steps into the future is your model able to model effectively and I mean it may just be that like the the temporal scales at which were actually having an effect on the planet don't match up to the temporal depth of the models that we're using to deal with so like you know the the the time of everyday phenomenology is kind of a cyclical kind of like 24-hour thing where like you'll plan maybe a few weeks or months or maybe years ahead but like there's a mismatch between that scale right like the scale of the generative models that we use effectively in everyday behavior and the scale of the effects that we're having on the environment so I think that the the you know the political decisions political decisions are made with like a two-year window at most right maybe four depending on the country but like you know it's little guarantee the next election you know it so of course there's like this huge incommensurate ility right like because the the the timescales of our practical can and the time scales at which the threats the existential threats to you know our continued existence as a species like that it's this in commence or ability that I think needs to be bridged yeah with any hope of you know yeah that's um that's a natural article that's where I'm cynical I think if anything like this rise of virtue signaling you know on the left and on the right it points to the fact that like you know the the the kind of disembodied ideal of a rational actor you know it isn't really what's at play here like people are I mean I think at was Isaac Asimov who said you know our fundamental predicament is that we have Paleolithic Minds medieval social institutions and godlike technology right like with our minds like our brains were optimized to handle you know it's Dunbar's number right it's an each of about 150 people timescales you know the point of culture in part the conservative power of culture is to obviate the need to think about long term regularities in any kind of detail I mean that's the point of constructing a niche you don't have to deal with time because everything that you need to deal with the temporality of the world is already pre encoded in our practices and in the physical layout of an environment so like the I see no reason to believe that like suddenly we're gonna start shifting our perspective if anything I think like you know we we've constructed like these these informational Ecology's that make us more resistant to actually you know challenging the depth like the temporal horizon at which we're engaging with phenomena and rather just like signal to our you know our people like us that we're on we're on the right side of history we're doing this right and they disagree because they're bad guys I think that's far more likely that you know I maybe maybe the promise of these meta Bayesian approaches that I was alluding to you miss to kind of sketch you know some paths or ins yeah actually like doing that without an you know infringing upon people's freedoms like if you just look at the US like they have as freedom-loving and I use that I say egg quotes because like I think they've got a certain view of freedom that I don't think everyone would agree with you know like you are free if you have a gun you know you are more free if you have you know you can bear arms or if you don't have your text as you'd have to pay as much in taxes because you know taxes of theft and I think I'll um be right yeah I just hope that like I think one of the most important things is as we said like taking our so we have evolved to exist in smaller tribes right but one of the wonderful things is we have developed this capacity to extract really useful information from the world and explain phenomena with it right back science and our challenge is to like take this you know godlike capability and use it with look I mean look just just to be a bit like self-critical so I'm not a climate scientist right I think I don't I don't annoy just a stop right so like when I when I say I believe like human climate change is real that's an act of faith on my part yeah right and so like yeah my suspicion is that that encapsulate like the human thought process like ultimately like there's something about like deferring to whatever is the default belief of the community that I feel that I belong to you that's like the main mode of operation even when we think that it yields results that we find are good right so you know I'm also part of this more like progressive anti-racist you know feminist and agenda but you know you might say that's I'm like this because those are the ideas that are available in my eco Miche I'm just you know conforming to whatever is so I think the the person who articulates this most beautifully is Joseph Henrich I really I really like I really really like Joe he's great he's uh he's based at Harvard he's like a an evolutionary biologist anthropologist and he works on the self domestication of humans and so he says well basically the the those very mechanisms that enable us to perform like these large-scale feats of incredible coordination right the things that make us like agreeable and able to cooperate with one another are also the things that like you know make us susceptible to fascism and to like authoritarian dictatorships were agreeable we just go with things you know like they're not the typical Nazi wasn't like a raging racist yeah you know they were just like quote-unquote good people just trying not to rock the boat and just going with whatever the and so you know that's really what like I think it's too easy for us to say like oh well clearly you know like we're on the right side of this because our beliefs happen to line up with whatever is the case in reality the cognitive operations that I go through to say like well you know I believe in man-made human-made climate change are essentially the same is like an empty box are going like oh well you know like I like I can't myself verify that this is the case mmm I don't know the science like ah my my my opinion rests on this division of labor but the yeah so I'm not particularly optimistic yeah look I've never thought about it in that way and you've just made me a little bit more pessimistic like I've never really thought about like that and we are like I just his face faith-based yeah yeah yeah because I have looked at them I like riotous assume the experts are experts in ER and just less trusted but everyone assumes that they're experts or experts right yeah that's the thing that's that's what's sharing a generative model is about but at the same time like there's like that's that's all that's true but then when you think about climate change when you think about that we are on this bounded planet like you know it's a closed system in many ways and that we are there's like eight billion of us a lot of us are using I think the value the number is like 30 times our basal metabolic rates you're like for us to live the lives that we live we need 30 times amount of energy our bodies needs like what do you think about that it kind of makes sense that things might be heating up you know like you don't need to like really but I'm not denying climate change no I I know you know I'm like I don't think you need to be like on this progressive side of things to actually just go through that line of reasoning do I critical ultimately the the big filter is natural selection right so we can have the beliefs that we want and we can have the generative models that we want but like if we're too off we're gonna die yeah and that's that's the bottleneck at the end of the day and that's also why I'm not super like optimistic because like the the way that these shifts happen typically is massive die-off yeah I mean like that's how generative models are not evolved a right the same way that everything else ends up involving like you know whether its cultural selection or natural selection like you know the maladaptive beliefs end up like causing negative consequences to their hammers and then not spread into the future and so like yeah I agree like you know I I think I personally I think I have good reason to believe in that make climate change and I also think I have good reason to think that if we don't take it seriously as a factual matter of affairs and there will be negative consequences but my fear is just that like even if I know this my motive justification is not fundamentally different and the one employed by people who believe in chemtrails and you know where I gave fog stuff like you know you heard Alex if you didn't watch that podcast with him Jurgen and no not the whole Robert I mean it's ridiculous it's ridiculous to fundamentally though like the difference is that I have I have more epistemic tools up my sleeve right I'm a scientist able I don't know I don't have the exact analytic tools that are used in climate science but I do I do know like a story out of anecdotes from you know rigorous they have kind of reviews of evidence but like fundamentally though like it's it's sort of about it's all about like the the kind of the so Joe Henrik calls them creds credibility enhancing displays you know it's it is that there's an economy of authority you know humans are the the only mammal species with a prestige system that's evolved independently of the dominant system right so like what was that system I just cut out a vestige okay prestige so Preston he's just a kind of freely conferred power right whereas like dominance is you know you listen to me because I'm gonna smack you if you don't write like prestige is the kind of thing that like you know Joe says the the the the top top climber at the base camp of the Himalaya right like everyone wants to climb with this person yeah because it just so happens she's the best right she knows how to you know go up the mountain she knows the path she knows what did you like the he's socially but it's it's like it's not that she's conferred yeah right but there's an economy of these of authority like in in almost every society there's a series of ritualized cues that indicate who is the knowing cast right so like you know scientists we're you know sorrows and doctors have their stethoscopes and teachers profs I mean less and less maybe like in the North American context but like you know profs will often dress up you know people in positions of power indicate not just power but like people in positions of epistemic Authority reliably indicate with like Ecology's of cues that they have this Authority means are just very sensitive to these cues you know we so it's it's you know it's it's what the Milgram experiments for example so I mean it's people deferred I mean I don't know if Milgram still works anymore wasn't good I think it's a bit dodgy it was yeah but anyway like there are these experiments from social psychology that tend to show that people just defer to whatever they think is a source of authority and I think that's more germane to human cognition than strenuous putting into question of whether the psychology of queues actually reflects something so I just think it's moral luck at the end of the day if you and I are on the right side of this kind of factual matter it's it's not it's not because the vinegar true that we cultivated we just happen to be in the right at the semuc meteor that allows us to make the right kind of inferences yeah but so my my fear though is that like making these kind of fact-based inferences takes a lot of energy and there are there are much fewer people who are able to make them than people who are like able to you know subscribe on YouTube and start a channel about chemtrails and whatever so like I mean on the whole I'm not I I don't know what's gonna happen I mean III don't see any reason to be optimistic to be blunt oh well we can do our best and see what happens I mean having said that like I I think that the kinds of tools that we're developing are giving us a better grip so for example like you know this this generate shared generative model business allows us to understand the Constitution right and the maintenance of epistemic communities in a robust way that might be amenable we hope to modeling into like experiment and so you know it it may be that like by studying the ways in which humans reason right and reason on the basis of like credibility enhancing displays right an arbitrary between different sources of epistemic Authority and so on it may be that like you know through that process mmm you know none of its age dependent just because I don't think we have much time it may be so yeah I someone that I really admire in the literature she's a theoretical neuroscientist and neuro imager she's with the neuro imaging department at King's College and she has this great model of how basically the the parameter range of your generative model becomes more and more rigid as you grow older and that's that's not because your brain is like efficient it's that it's adaptive I mean presumably when you're middle-aged you should assuming like a relatively stable structure in the environment you should have less to learn than someone who's like half your age you know that there is a rigidify of the generative models that's age dependent I mean my own my own fear has to do with like the the rapid generational shift so like you know the boomers that gets so mulling maligned in the you know just in the media you know just abused and everything will be they grew up in another world yeah we're like you know journalism was you know like everyone had the same everyone had the same informational environment right liking right you just wasn't precisely because there were like three sources of information right so yeah so whereas people I don't know exactly how you are under 30 and so like I I grew up with the internet and with the assumption actually like part of what it is to have a valid epistemic process is to bet you know the source of the information where you're getting this whereas like if you grew up in that it's all to it it was like you look at something and it's like this is oh yeah but exactly but if you grew up in an ecology where there are three sources and information you know like the the news at five you know your local journal and whatever your big-city journal is like then then your relation to sources of information and the very process by which you bet your own beliefs is going to be radically different yeah so it may be what we're seeing now maybe like I mean you know and I don't want to make it only into a generational thing I mean there are a young conspiracy theorists but it has to do with like you know exploiting the basic kind of you know mechanisms for attributing epistemic Authority I mean you know you'll notice like Trump does this all the time you know like he prefaced is his sentences with material that would enhance his credibility in certain communities right so saying like you know I'm the best at Nenana no one understands X better than I do like the these these cues work in a community were typically the people who say this are the people who know right now so if you grew up and you were uncultured in a community where you know people who signal in this way typically are right about things and it's not surprising that you'll end up like you know adhering for this and I don't only want to make it into a right-wing thing I mean I think oh it's on the both sides for sure for sure it's it's just we live in such interesting times you know Michael I think we're just see right now is self-righteousness self-righteousness yeah on everyone's side I think yeah it's just you know from a philosophical point of view I come from this tradition called hermeneutics so I mean my favorite philosopher is uh this-this-this philosopher called gadamer yeah gadamer's philosophy is a philosophy of interpretation so hermeneutics is sort of a corrective to phenomenology phenomenology says let's start from the first person lived experience and its immediacy hermeneutics says there's no such thing as immediate first-person experience all experience is all always already mediated by structures of interpretation but anyway I don't want to dwell on that but I say this because gadamer says beautifully somewhere like the starting point of my philosophy is that my interlocutor could be correct right like I might be wrong it's about interpretation and finding a common interpretation and I think that's been completely evacuated like the discussion is about showing the other person why they are wrong and showing people on our side why we are so virtuous right like the you know we're really going to stick it to the lids or you know just like finger-pointing and calling people or whatever isn't as hot today like it's like it's it's only it's only epistemic this talk that we're having this discussion is only epistemic for show it's not really about epistemic it's about confirming what I believe right and so so telling you who disagree with me that you have to be wrong ultimately right and yeah it's it's it's and about showing you know others that you know we're we're also valid epistemic agents you know I guess like if that makes me think and you have to thank you for a while and you know I'm just echoing what I've heard a lot of other people say but humility from you know a scientific standpoint from the standpoint of policy but also in our own interactions with each other is has never been more important right like that we can learn from the person that we're talking with or arguing with yeah so that our actions like we actually don't know what may happen in the world we don't know how bad the current of Irish could be we don't know whether or not our climate change models are actually correct and Ravin that we do not know we need to be carried to be extra careful because things could go better than we think but things could also go a lot worse yeah precisely and you know we have to be open to and again I'm not saying that we have to listen to these crazy denialists who you know cherry-pick their data and who are obviously not like engaging in the same kind of scientific process that we are but like I especially mean like moral certainties you know I mean it may be that the way that we see the world even even our our you know most deeply cherished values maybe they're off you know to a certain extent and and I think that it's the it's the self-righteousness of everyone involved in these political conversations that makes it and you know I'm a many minds about this because on the same time you know at the same time if someone's position is that like we should exterminate whatever you know a group of people because of whatever you know reason and that also doesn't float you know like I I don't think that we so it's it's it's all sorts of complicated and I don't have any other dancers I just look at the problem honestly yeah well that's the first step to solving it right mmm yeah understanding it I mean the one thing we didn't discuss is is what the approach can't do so yeah there's good reason to think that nature you know life if life exists right if life is a useful concept if we don't eliminate it from our ontology there's a good reason to think that life isn't a non-equilibrium steady state process it isn't it isn't like the so like evolution isn't going anywhere right if life is this open-ended kind of thing that's like a tree yeah exactly yeah and and and it works on a process of drift which is not predictable you know like so I mean natural selection works there is variation selection retention right so like that process is not so the the technical term is ergodic so basically everybody mean yeah average yeah yeah yeah okay so like the so you know like so if you think of the space of all possible states of the system like an ergodic system means that there's some density towards which the system eventually converges right such that on average so that the average of your measures will converge to a measure of your average yeah because there is a density at the end of the day right okay there's reason to think and in fact like I'm working with people who are showing this experimental II there's reason to think that at several relevant scales there is a non-equilibrium steady state density one of the easy demonstrations of this is that you I assume I mean maybe you have a different physiology for climbing but like you're roughly the same height from day to day right your your your blood pressure is roughly in the same or in the same neighborhood you know you're all that you have these these states that you revisit frequently you know the such that you can say that there's something like ass phenotype all right that you returned to characteristically but evolution itself is an open-ended process that might not have such a like an endpoint right so so like right now the the formulation is in terms of non-equilibrium steady state systems but it doesn't extend to non equilibrium non steady state systems so so the ways right given the time scale we're not non steady it's really like it depends on your temporal scale in scale at which this doesn't apply yeah yeah yeah and in fact like it may be that like the the more socially interesting stuff that you're interested in can't interests by this framework for precisely that reason what basically if if the attractor doesn't exist yet there's nothing you can do from variational e like there's nothing to converge to mmm well sort of right I agree there's no there's no point of convergence however one tenant that you could say is the need to survive right and a part of that diversity of approaches so well so you can still get something out of it so you can say like there are like you can say there are something like different attractors that are competing right so there's the liberal the progressive and the conservative kind of attractors those sort of competing for where the overall trajectory but you might say that but look like what I want to say is something like well you can use the current formulation to model something like feudal friends but not France during like the Revolution and the terror when they're there just isn't a clear attractor like so the entire approach rests on being able to write down this non-equilibrium steady-state density like this the phenotype if you're able to characterize the phenotype then you gave us this effect so that's their problem right like life is a process it's not discrete it's like so you're constantly becoming I mean and I I didn't the way I think about life and death more specifically is that we don't really die like we like life is just about information creation consolidation and propagation and we are just apart we're just passing the baton right like we live on through our genetic code but also through the culture the ideas possessing I couldn't agree more yeah yeah all I'm saying is that like you know and and as an advocate of the approach work I also want to hope that there's a way to extend the the formalism to non steady state systems but if you're asking me like what can't this do because it does seem like it can do almost everything well so what Kant did do the historical change mmm can't do it right now like III for principled reasons this isn't like an accident but like so you know the approach hasn't yet been applied to the formation of epistemic communities that's not a principled limitation it's just that it hasn't been done yet yeah so that's what we're doing well if this is hammer and there's so many nails now right exactly but like this may be a principle limitation it may be that this kind of approach can is only conceivable for non-equilibrium steady-state system and certainly the current mathematical formulation even in Carl's like totally integrative 20/20 like monograph that really contains like the whole approach even there like it's formulated in terms of non equilibrium steady state so it may be that like and this is what's frustrating for me as a scientist like maybe the most important questions are not can't be addressed in this kind of framework because at least the most ethically important questions may be those have to do with when the attractor hasn't consolidated maybe I don't know or if there even is one right like it like that's I think I've on the promise of like ethics in general it's like well is there an objective good and like what are we moving towards and like I think in a way it's kind of the wrong question I don't think there is an objective good like if you take it fueling materials the materialist approach like we have just like minimizing entropy locally and trying to do the best as possible as we move up scales and when I say we I mean life in general right and that morality is emergent and like I think like obviously we'd say that like that which is more complex so that you know like life is good like if you think of the world with life or no life we'd say that life is good we're obviously biased but I don't think there's anything wrong with like to say there's this that there's no like if you take this perspective that we have just kind of emerged from soup and morality has come with it and that there may I believe that there would be things that are that all forms of life intelligent life would think would deemed to be of moral worth so you can think of that as objectively moral but like to see there as being objective morality extricated from life I just I don't think that makes any sense you know I I agree and that's consonant with the framework that we're too since from from our vantage point morality is just encoded in a specific kind of shared generative model you know so so my question so let's say there's two worlds right and I know that I have to choose between one and I've got a flourishing ecosystem bamas only rainforest or I've got a shitty one like a desert right I just say one of them has to exist which one would exist everyone would choose the Amazon rainforest one not everyone if you're a Bedouin you'll prefer the desert yeah but I think like they are I think on average and I think that is important right because it depends on what you've like you're adapted to like I think that we are like we have moral intuitions that point us in the direction of that which promotes order and survival and the increasing and here's the rub like that is defined in terms of the non-equilibrium steady-state density so you're still not out of the woods it's not objective like the the good is just defined in terms of like and like our body's adapted this so I'm a Canadian like you know when it's five degrees Celsius outside I can go outside in a t-shirt and I'm totally fine like but I don't handle the heat well like when I went to Australia Jesus Christ like I went in February and Sydney just kicked my ass like I just can't handle this 40 degree heat so I lived in Malaysia right and I'm like you I can I can handle a cult but the heat just ruins me so I had a sweating problem and I just realized I was just in their own country well but your body does adapt I mean like after several months of this you start sweating less and that's what I mean though is like the third you know it's all about the complete package of priors that you embody so there really isn't the way to say well this is good and this is bad I mean you know think of like a preference for spicy foods I mean it's even or sad Oh masochism as a sex thing I mean it's it's just it's just the case that like you know there's no objective kind of mapping it's just whatever is you know like whatever is available and whatever can lead to like a targeted configuration I mean the the bit that I kind of skipped over for the social stuff is that like what's sharing a generative model allows you to do at the end since you're sharing all your expectations is to arrive at the same target configuration so all of these multi scale models they rest on the idea that a bunch of components sharing a generative model if they engage in individual inference they can still reach this kind of target configuration so that's how all of this morphogenesis cellular differentiation stuff inactive in family and when I met Carl what made my nose bleed initially was him telling me you know the exact same mathematics that we use in the morphogenesis stuff you could use in the social stuff because it's it's it's sharing a generative model it's just that in one case the generative model is a metaphor for a shared genetic and epigenetic code like when what you're modeling is like cellular differentiation the genetic code is a shared generative model that tells every cell well what kind of thing should you expect sentence given the kind of cell that you are right so the the same the very same mathematical framework applies to social dynamics it's just that rather than you know what kind of you know intracellular signalling you're expecting it's what kind of social cues do I expect to see right but it's the same idea yeah yeah why is it a target configuration by sharing a generative mall and the twist for the whole multiscale story is that like the this can be reiterated infinitely so your cells share expectations that cell networks share expectations organ networks you know organisms yeah mall or social groups like the family unit larger social groups like a like a community like all of these can be cast as like successive recursively nested of a system that at each level share the same generative model and they're all doing the same thing example is a cell that doesn't write it was like a cancer cell doesn't have the same generative model as your other cells because it's genetic material has been altered you know so that's really the perspective from which I'm coming to understand like it's also why I'm so pessimistic to be honest with you I mean the thing that you see is that you need to share a generative model to for there to be a target configuration right like the the the share generative model is what induces the non-equilibrium steady state at the superordinate level the fact that we all share the same generative model means that our our behavior will look as if it's converging toward some kind of phenotype at the high dose that's how you get cool like self-organization that scales like that exactly okay that's what dropping me drew me into the free energy principle initially so we need that I just static global like I had a planetary and I own now yeah but then the question is can we construct the generative model that's both complex enough to like capture this kind of you know that the model is complex enough to capture the complexity of the phenomenon but also transmissible you know like one of the big problems that we're having right now in the approach is that like this all rests on like this high-flying one of the things that people don't get about this is that I get builds on quantum mechanics it builds on classical mechanics it builds on thermodynamics so it's much more complicated than quantum mechanics so like you know there's there's a transmission issue that gets in the way at some point because like not everyone can be expected to like assimilate that level look like that's why I wanted to have this conversation because like I'm I can like read stuff and understand it like generally like looks like I'm the genius but I sometimes I can read stuff and sort of get picture but this stuff is a bit impenetrable like I don't have a mathematics mathematical background and yeah I mean unfortunately like the math is required yeah it's not like it's not like a gloss it's that like it's sort of like look if I ask you you know bees make honey combs right yeah why are honey combs shaped the way they are as strength and space-filling yes precisely it's space-filling it's that hexagons are like one of the one of the geometrically optimal ways to tessellate or to completely cover a two-dimensional space so in this case it's a bit of math that explains the behavior is that like you need to appeal to the mathematical fact that heggs hexagons can completely tessellate a 2d surface and this can be generalized into three dimensions right like you need that piece of math to explain the behavior of these so what asks where I need to know right here I mean the only way to do that is to like really slowly go through the literature I mean I've been doing this like I've just been doing active in different stuff basically since 2014 and I'm still learning but I mean like you know the really it's a multidisciplinary approach like there are people from computer science there are people from physics there are people from philosophy and at from anthropology and like the full picture is bigger than any any one person with their tools set can address well max well I think I've taken a decent chunk of your time today sir they carry in your f oh yeah yeah it's been lots of fun um if people want to find out about your work keep up to date with it where should they go online how can they find you online well I mean I'm on research gay and academia.edu but all of my research basically yeah essentially everything I've ever written is available either as a preprint or open access actually the vast majority my publications are available up in access so I like to say that my hockey card is my Google Scholar account yeah yeah really I like that I like a Twitter account if you want i i'm also on twitter i Twitter's my public kind of academic thing so like I keep I keep everyone up to date like when this is released all OH I'll advertise it on my Twitter so yeah Twitter is the best way to keep like in the loop but like all of my work is accessible on my Google Scholar account like you'll find links to all of the manuscripts for free so has been wonderful okay that's been good cuz I mean is research for this right it's nice actually yes I mean you know we we we we are definitely not in favor of paywalls and stuff so yeah all of my stuff is accessible so yeah yeah thank you for thank you for the invitation honestly Sam this has been very very enjoyable yeah that's a good fun it's been good fun thanks for taking the time and uh yeah I'm sure we'll do it again sometime soon yeah I'd love to and whenever you want me back it was really a pleasure so oh so uh yeah let's do it again